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Does history matter? Colonial education investments in India[1]

By LATIKA CHAUDHARY and MANUJ GARG*

Can good policy overcome or alter the effects of history? This question is addressed in this article using unique district-level data for 148 districts of former British India. Controlling for observable differences in geography and income, the ordinary least squares estimates suggest a large and positive effect of colonial expenditures on rural primary education in 1911 on rural literacy up to 1991. However, instrumental variable estimates that control for the endogeneity of colonial investments suggest that the effects of historical spending are significant only up to 1971. Two policy changes can account for these findings: an increase in spending following the 1968 National Education Policy and a greater emphasis on the universal provision of public goods such as schools in the 1970s. Unlike recent studies documenting the persistent effects of historical investments on contemporary outcomes, this study emphasizes how effective policies can overturn the effects of history.

Over the course of the twentieth century, many developing countries have grappled with education policies to increase both access to and quality of schooling for their citizens. India is no exception. In the first two decades after independence the Indian government increased education expenditures, which had been very low under colonial rule. However, the early focus of education policy was on developing a technical secondary education system comparable to those of developed countries. The cause of primary education was more seriously addressed in the National Education Policy of 1968, which called for increasing public expenditure on education to 6 per cent of gross national product (GNP), along with providing free and compulsory primary education. Further recommendations along these lines continued in the 1970s and led to a more extensive National Education Policy in 1986.

Although policy recommendations to improve schooling have continued at a brisk pace over the last two decades, it is easy to judge Indian education policy harshly. India is still a long way from achieving universal literacy, with striking differences persisting across regions, gender, caste, and religion.[2] However, contemporary debates on the merits of the Indian education system often forget that

*Author Affiliations: Latika Chaudhary, Scripps College.

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[2] Many researchers have written about the wide differences in literacy within India and the problems with Indian education policy. For example, see Choudhury, 'Inter-state and intra-state'; Drèze and Sen, *India*; Probe Team, *Public report*; Drèze and Kingdon, 'School participation'; Pal and Ghosh, 'Elite dominance'; Tilak, 'Five decades'; idem, 'Public subsidies'.

India inherited a poorly funded public education system and many of the regional differences have long historical roots. For example, as early as 1911, literacy levels in western parts of India, known then as Bombay presidency, were almost twice as high as in eastern parts of the country such as Bihar and Orissa (6.1 as compared to 4.1 per cent).[3] The southern princely states of Cochin and Travancore, the current state of Kerala, were leaders in educational attainment, as they are today.[4] Similar patterns are discernible at more local levels: Ahmedabad district in western India was among the five most literate districts of British India in 1911 and continued to enjoy this position until 1991. Such patterns are remarkable given the many changes experienced by India throughout the twentieth century and the major public investment in education in the second half of the century.

Universal literacy is an important goal, but a first step towards achieving such goals is to break from historical patterns that perpetuate regional differences. Policy prescriptions to improve education can be very different depending on whether historical factors are impeding progress or whether contemporary factors are at work. In this article, we use district-level data for 148 districts of British India to test whether colonial investments in education continued to influence post-independence literacy from 1961 to 1991. We focus on one particular measure of colonial investment, local district board expenditures: a major source of public funding for rural primary education under the Raj.

Colonial investments were not randomly assigned across districts and so public spending in 1911 may be correlated with historical factors that influenced subsequent education. To address this concern, the cross-sectional analysis controls for historical income, development, and social structure. The ordinary least squares (OLS) coefficients suggest a large and positive effect of colonial investments on rural literacy and rural primary school graduates: a 10 per cent increase in 1911 public spending per head of rural population is associated with a 7.7 percentage point increase in rural literacy for the population aged 10 to 19 in 1961, which decreases to 6.5 percentage points by 1991. We find no significant effect of colonial spending on control outcomes such as the number of secondary school graduates. Our measure of historical investments was targeted at primary education, so in theory we should not find effects on secondary education.

However, the OLS estimates may still be biased if colonial policy involved heavier investment in districts with a strong private demand for education that is not captured by the controls. Hence, we instrument for 1911 public spending using 1911 land tax revenues. According to historical accounts, subjective forces unrelated to education created quasi-random variation in revenue assessments conditional on observable differences in geography and development across districts.[5] In support of these accounts, we find no significant correlation between 1911 land tax revenues and the proportion of the rural population engaged in commerce in 1961–91, a variable strongly correlated with literacy levels in 1961–91. This test suggests that historical revenues are not systematically related to contemporary factors that influence the demand for education.

Unlike the OLS estimates, we find the instrumental variable estimates on 1911 public spending are positive and significant up to 1971. Colonial spending in 1911

[3] Chaudhary, 'Land revenues', p. 185.
[4] Ramachandran, 'Kerala's development', highlights the historical role of the state in Kerala's success story.
[5] Chand, *Some aspects*.

is statistically unrelated to both literacy and the number of graduates by 1991. This suggests the new priorities of the 1968 National Education Policy, along with other policy changes in the 1970s, were successful in de-coupling current outcomes from historical investments. In the 1950s and 1960s, public education allocations were based on historical patterns that perpetuated differences in spending, inherited from the colonial era.[6] From the 1970s, the universal provision of public goods became an important policy goal and public funding for education also increased. Subsequently, there was an increase in primary schools and other public goods, especially in previously underserved areas.[7] While the state may have failed to achieve universal literacy, these policies were at least successful in breaking historical ties.

This article contributes to a large and growing literature on the long-run effects of history on contemporary outcomes.[8] While early works established the importance of history for modern economic growth, they were unable to uncover the deeper pathways connecting history to modern outcomes, unlike more recent studies that focus on individual countries. For example, Banerjee and Iyer study land tenure relationships in colonial India and show that areas where land collection rights were invested in landlords have performed poorly on a variety of agricultural and investment outcomes, compared to areas where such rights were invested directly with cultivators.[9] Huillery finds large and persistent effects of colonial investment in education and health in the early twentieth century on enrolment and stunting rates in the 1990s in the former colonies of French West Africa.[10] A high degree of persistence in public investment across the colonial and post-independence periods is responsible for these patterns.

Dell finds large, negative, and persistent effects of the *mita* forced mining system established in Bolivia and Peru between the late sixteenth and the early nineteenth century on current consumption levels.[11] Differences in public goods such as infrastructure are important drivers of the results. However, similar to our story, she also finds the effects were not as persistent for education, as a result of a strong universal system of schooling adopted in Peru. Thus, effective policies can reverse or counteract the effects of history. The effects of colonial investment in India were relatively short-lived because of the existence of a strong central government that was focused on increasing public goods such as primary schools.

The rest of the article is organized as follows. Section I reviews the institutional history of the provision of schooling. Section II describes the data. Section III discusses the OLS results. Section IV highlights the endogeneity problem and presents the instrumental variable results, and Section V concludes.

[6] See Tilak, 'Center-state relations', for details on education financing after 1947.

[7] Banerjee and Somanathan, 'Political economy'.

[8] See Nunn, 'Importance of history', for an excellent review of this literature. Seminal works include Acemoglu, Johnson, and Robinson, 'Colonial origins', and Engerman and Sokoloff, 'Factor endowments'. Both studies focus on the role of institutions in explaining cross-country differences in economic development. Acemoglu et al. focus on disease environment, arguing that colonies with high settler mortality (on account of the unfavourable climate) led to colonizers establishing more extractive institutions with poor protection of private property rights that have persisted and have led to poor economic development. In contrast, Engerman and Sokoloff argue that a geographical predisposition for crops such as sugarcane, grown more commonly under plantation style economies, created significant differences in economic and political inequality, leading to more unequal institutions and worse economic outcomes in parts of the Americas.

[9] Banerjee and Iyer, 'History'.

[10] Huillery, 'History matters'.

[11] Dell, 'Persistent effects'.

I

Colonial investment in education

The provision of schooling was not an important priority for the Raj, and public spending in India remained low up to independence.[12] The British Crown via the Government of India controlled education policy from 1858 to 1919, with decentralization of public schooling to provincial governments occurring in the 1870s. Another wave of decentralization in the 1880s devolved the provision of primary schooling to rural district and urban municipal boards.

We use public education expenditures by rural district boards in 1911 as our measure of colonial investment. These data are suitable for three reasons. First, information on district board spending is reported for a majority of British Indian districts, unlike data on public spending by provincial governments and urban boards, which do not have the same coverage. Second, these investments represent the majority of public spending on primary education, as we show below. Third, we believe the 1911 series are representative of the regional variation in public spending over the colonial period. The correlation between expenditures in 1911 and those in 1901 and 1921 are very high for districts for which we have comparable data.[13] While there were policy shifts after 1911, such as the introduction of compulsory education in 1918 (initially limited to urban areas) and an increase in public spending in the 1920s, the differences in public spending across regions remained the same.[14]

District boards were set up in the early 1880s following the 1882 Resolution of Local Self-Government, which called for the establishment of rural district boards, sub-district boards (where possible), and urban municipalities with up to a two-thirds majority of non-official Indian members that were either elected or nominated by British district officers. Despite the stated importance of electing Indian members, board chairmen were often British officials, assisted by members of the landed Indian elites and upper castes.[15] These rural boards were entrusted with the provision of primary education, local infrastructure, and medical services, with 80 per cent of expenditures allocated between schooling and infrastructure.

Within primary education the boards served two key functions. They managed a few high-quality schools directly, and provided public subsidies to schools that were privately managed (known as aided schools). All the money was distributed in rural areas where almost 90 per cent of the population lived. Chaudhary finds district board spending on education had large and significant effects on male literacy in the early twentieth century, but not on female literacy.[16] Table 1 reports district board expenditures as a percentage of total education spending, primary school spending, and public primary school spending in 1912–13 for British India

[12] Chaudhary, 'Caste, colonialism and schooling'.

[13] The correlation between 1901 and 1911 education expenditures by rural district boards is 0.9. These data cover over 100 districts of British India. In 1921 we only have the data for Bombay presidency, and the correlation between 1911 and 1921 education expenditures is 0.89.

[14] As noted in Chaudhary, 'Land revenues', similar to the late nineteenth century Bombay was the leader in education spending per capita in 1931, while Bihar and Orissa province was at the bottom. There is no reason to suspect that the within-province differences did not follow the same pattern.

[15] Tinker, *Foundations*.

[16] Chaudhary, 'Taxation'.

Table 1. *District board revenues and education expenditures*

	District board education expenditures, 1912–13			
	Per 1,000 of population (£)	*As % of total education expenditure*	*As % of total primary education expenditures*	*As % of total public primary education expenditures*
British India	3.59	15	49	65
Bengal	2.43	8	40	80
Bihar and Orissa	1.65	13	34	61
Bombay	10.27	20	42	59
Central Provinces and Berar	4.58	25	61	66
Madras	3.03	12	34	38
Punjab	5.12	18	85	93
United Provinces	3.55	21	93	100

	District board revenues per 1,000 of population (£), 1912–13			
	Cesses (additional tax on land revenues)	*Other sources (for example, provincial grants)*	*Total income*	*% cesses and other sources to total income*
British India	6.53	5.96	16.65	75
Bengal	4.28	4.35	10.51	82
Bihar and Orissa	4.56	3.08	9.66	79
Bombay	10.84	10.23	25.78	82
Central Provinces and Berar	4.99	9.34	16.07	89
Madras	10.80	10.13	31.85	66
Punjab	10.05	6.75	22.92	73
United Provinces	5.20	4.26	11.53	82

Notes: The statistical abstracts separate education expenditures into direct expenditures on different levels of education and indirect expenditures on buildings and so on. In the calculations, we constructed total primary education expenditures as the sum of direct expenditures on primary schools and a proportion of indirect expenditures on buildings, with the proportion equal to direct primary school expenditures relative to total direct expenditures on education. In the bottom panel, provincial contributions or grants to district boards are included under 'other sources' along with school fees and miscellaneous income sources. The omitted category of income is tolls.
Source: Chaudhary, 'Taxation'.

and the larger provinces. District board spending represented 15 per cent of total education spending, 49 per cent of primary education spending, and 65 per cent of public spending on primary education.

Public spending, at £3.5 per 1,000 people, was low and translated into less than 0.5 per cent of per capita GDP in 1912–13.[17] Although it increased in subsequent decades, spending remained under 1 per cent of per capita GDP until independence. Table 1 also highlights the vast differences between provinces. Bombay, the highest-spending province, outspent Bihar and Orrisa, the lowest-spending province, by a factor of six. There was significant variation within provinces as well. For example, in Bihar and Orissa, spending per capita in 1911 ranged from 0.015 to 0.049 rupees.

Land tax revenue and the type of land settlement (permanent or temporary) were the major drivers of provincial differences in public spending.[18] The land cess, an additional tax on the existing land revenue, was a key source of revenue,

[17] Public spending was low in India compared to other countries such as Brazil, Russia, and China. See Chaudhary, Musacchio, Nafziger, and Yan, 'Big BRICs'.
[18] Chaudhary, 'Land revenues'.

contributing 39 per cent on average but with significant differences across provinces (see the second panel of table 1). The tax rate for the cess differed by province, but was generally uniform across districts within the province. In addition, the rural boards had no financial autonomy to raise revenues by increasing tax rates. Provincial grants to district boards accounted for another 25 per cent of board income, while tolls on roads and ferries contributed 10 per cent. Finally, school fees, income from cattle pounds, and private contributions made up the rest.

On account of the permanent settlement in Bengal and Bihar, land tax revenue had been fixed in cash in 1793 and revenue in these areas did not increase with changes in prices and agricultural productivity. Unsurprisingly, these provinces also had less grant money to transfer to their districts. In temporary settlement provinces (the rest of British India), land tax revenue was revised every two to three decades to incorporate changes in agriculture, albeit with a substantial lag. These areas had more public money and hence could distribute larger grants to their districts. The qualitative evidence suggests that grants within provinces were distributed on a per capita basis, with a stated preference for favouring poorer districts. It is unclear whether this redistribution policy was followed in practice. The individual preferences of district boards may also have influenced the distribution of funds to education versus other local services.

For the purposes of this article, it would be useful to establish whether districts receiving larger public education funds were positively or negatively selected. Given the many differences between the historical provinces and post-independence states, we exploit within-state variation in the analysis by including state fixed effects. The institutional history provides some guidance on the across-province variation (land settlement and land tax revenues), but there is limited discussion of within-province differences, despite their significance. If the boards were distributing public funds in response to private demand for education, then we would expect more developed districts to receive more money. However, if boards were targeting districts with fewer schools then we would expect less developed districts and perhaps those with larger disadvantaged populations to receive more money.

We know that districts with a larger *brahman* population (or upper castes more broadly defined) had more schools, as did those with more Christians, on account of the missionaries. However, districts with more caste and religious diversity also had fewer primary schools.[19] Since the land settlement system was the same within provinces, caste composition and social structure were perhaps more relevant to differences within provinces. We discuss the endogeneity of colonial investments in more detail in section IV, where we propose an instrumental variables solution drawing on the land tax revenues in each district.

Education policy in independent India

The most significant change in policy after independence was the sharp increase in public spending on education. Immediately after independence, the government-appointed Kher Committee (1948–9) recommended an increase in

[19] Chaudhary, 'Determinants'.

Table 2. *Total education expenditures per capita (nominal), rupees*

	British India				*Indian Union*				
	1901–2	*1911–12*	*1921–2*	*1931–2*	*1951–2*	*1961–2*	*1971–2*	*1981–2*	*1987–8*
North	0.11	0.16	0.71	0.92	3.01	7.27	18.69	46.91	70.71
South	0.21	0.23	0.80	1.21	3.94	9.96	24.62	52.46	87.04
East	0.16	0.19	0.55	0.68	2.83	7.38	16.62	40.26	72.22
West	0.31	0.40	1.53	1.83	5.62	12.89	32.56	73.92	107.60
India	**0.17**	**0.21**	**0.74**	**1.00**	**3.49**	**9.02**	**22.57**	**52.57**	**82.91**

Notes and sources: Data for 1901–2 are from House of Commons, *Review of Progress of Education in India, 1897–98 to 1901–02. Fourth Quinquennial Review (East India: Education)* (P.P. LXV.1, 1904); data for 1911–12 are from House of Commons, *Progress of Education in India, 1907–12. Sixth Quinquennial Review* (P.P. LXII.1, 1914); data for 1921–2 are from Richey, *Progress of Education in India*; and data for 1931–2 are from Government of India and Anderson, *Progress of Education in India, 1927–32: Tenth Quinquennial Review*. Data from 1951 to 1987 are from Government of India, *Statistical Abstract of the Indian Union*. The pre-1947 data for British India include parts of present-day Pakistan and Bangladesh and exclude the princely states, while the post-1947 data are for the entire Indian Union excluding Pakistan and Bangladesh. In the colonial period (1901–31), 'north' includes Punjab, United Provinces, and Central Provinces; 'south' includes Madras presidency; 'east' includes Assam, Bengal, Bihar, and Orissa; and 'west' includes Bombay presidency. After independence, 'north' includes Jammu and Kashmir, Himachal Pradesh, Haryana, Punjab, Rajasthan, Madhya Pradesh, and Uttar Pradesh; 'south' includes Andhra Pradesh, Karnataka, Kerala, and Tamil Nadu; 'east' includes Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Tripura, Sikkim, Bihar, Orissa, and West Bengal; and 'west' includes Gujarat and Maharashtra. The north-eastern states in the 'east' region are included when their data are individually reported.

public expenditures coming from different levels of government.[20] The initial emphasis, however, was on technical and secondary education, and it appears that primary education needs were ignored. Due to the limited nature of progress in the 1950s, another commission, the Kothari Commission (1964–6), called for more public expenditures with a goal of reaching 6 per cent of GNP by 1986. Unfortunately, the 6 per cent target has remained elusive, despite renewed calls for meeting this target in the 1986 National Education Policy and in policies passed over the last two decades.[21] That said, India has made important strides in increasing access by allocating more public funds to education.

Public expenditures on education increased from 0.68 per cent of GNP in 1950–1 to 4.1 per cent of GNP in 1990–1.[22] In nominal terms, total per capita expenditures increased from almost 0.2 rupees in 1901–2 to almost 83 rupees in 1987–8, right after the last National Education Policy (table 2). In another break from the colonial era, most of the increase has come from public funds and the composition of expenditure has moved overwhelmingly to the public sector. In 1901–2, total state funding accounted for 48 per cent of expenditure on education in British India. By 1981, total state funding accounted for 85 per cent of total expenditure on education.[23]

Table 2 also reports per capita education expenditures by region over the course of the twentieth century. Before we describe the patterns, an important caveat should be noted. The data series in this table do not follow the exact states and provinces in each region because of changes in boundaries. The data for the

[20] Initially, education was a state subject; in other words, state governments financed the majority of expenditures. However, in 1976 it was moved to the concurrent list, giving the central and state governments joint authority.

[21] Tilak, 'Kothari commission'.

[22] Shariff and Ghosh, 'Indian education'.

[23] These calculations are based on data from Government of India, *Progress of Education*. The composition of spending in 1981 is from Tilak, 'Center-state relations', tab. 1.

colonial period are for British India and include areas in present-day Bangladesh and Pakistan that were part of British India. On the other hand, the post-1947 data exclude Bangladesh and Pakistan, but include the princely states that were incorporated into the Indian Union in 1947. Hence, we must be cautious about drawing strong conclusions from this table.

Apart from the marked increase in total expenditures, there is strong regional persistence that suggests that post-independence public investments may have built on colonial investments. In each decade, the southern and western states have enjoyed high levels of expenditures on education compared to the northern and eastern states. However, the differences are starker for the colonial period compared to more recent decades. For example, the western region outspent the lowest-spending region (the east) by a factor of almost three in every decade in the first half of the twentieth century. However, by 1987–8 the west was outspending the lowest-spending region (the north) by a factor of one-and-a-half. Although it is difficult to construct a comparable series for primary education spending, it would appear the differences would be more muted for the post-independence period because states with lower spending tend to allocate a larger proportion of their budget to primary education.[24] In 1991–2, the northern and eastern states allocated 53 per cent of their budget to primary education, compared to 48 and 46 per cent by the southern and western states respectively.

It is difficult to make strong predictions about whether we expect colonial investments to influence post-independence outcomes. On the one hand we observe strong persistence in spending, but on the other hand we also see that the inequality between high- and low-spending regions declines over time. The major policy shift seems to have occurred after the 1968 National Education Policy, following which education expenditures increased to 2.3 per cent of GNP in 1970–1 and then to 3 per cent in 1980–1. This matches accounts of an effective intervention by the central government in the 1970s to increase the provision of public goods.

Using constituency-level data, Banerjee and Somanathan find strong convergence in the provision of public goods between 1971 and 1991.[25] Electoral constituencies with a larger proportion of villages with primary schools in 1971 seemed to experience significantly slower growth in the subsequent 20-year period, when controlling for unobservable changes at the constituency level. Interestingly, the coefficient on the 1971 level for primary schools is of the largest magnitude and statistically significant. Banerjee and Somanathan attribute their findings to explicit mandates made by the Indian government to increase public goods. They also find that access to public goods improved for some marginalized groups such as the Scheduled Castes compared to others such as the Scheduled Tribes. Although the Indian government has affirmative action reservation programmes to increase the political and economic representation of these groups, Banerjee and Somanathan suggest that Scheduled Caste areas were more successful in improving access to public goods because of their growing political clout. This would also suggest another break from the past, when these groups had limited access to education.

[24] Shariff and Ghosh, 'Indian education'.
[25] Banerjee and Somanathan, 'Political economy'.

We turn next to describe the data we use to investigate whether increased spending in the decades after independence and the associated policy changes weakened or exacerbated the relationship between colonial investment and contemporary outcomes.

II

For the analysis we construct a historical dataset of 148 districts of former British India in the current Indian states of Andhra Pradesh, Bihar, Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Orissa, Punjab, Tamil Nadu, Uttar Pradesh, and West Bengal (see figure 1). The dataset merges information from a variety of sources including the colonial censuses, post-independence censuses, historical district gazetteers, and the India agricultural and climate dataset.[26]

The provinces of British India do not correspond to post-independence states, but district boundaries have remained relatively stable over time.[27] Using the *Administrative Atlas*, we merge district-level information from 1911 to post-independence outcomes for the same district.[28] In the case of larger districts that were carved into multiple smaller districts over time, we aggregate the data to correspond to the boundary of the older district. In a few instances where there was a substantial transfer of territory between two or three neighbouring districts, we aggregate the data for the relevant districts (historical and contemporary) to construct a 'super district' that does not correspond to a specific administrative unit. While far from ideal, this allows us to compare the same geographic area over time.

We use the 1911 colonial census to construct measures of development and social structure that may have jointly influenced colonial investments and subsequent literacy.[29] Caste composition affected schooling via both the individual preferences of groups and the level of social diversity. For example, education had a strong community element and merchant groups such as Jains had high levels of literacy because literacy was a useful business skill. Districts with more *brahmans*, the traditional priestly caste, had a private demand for education and the means to lobby public resources in their favour. To control for these differences, we construct the 1911 population share of *brahmans*, Muslims, Christians, lower castes, and tribal groups. In addition, we construct a Herfindahl index of caste and religious fragmentation.[30] Such measures of social diversity have been negatively correlated with the provision of schooling in colonial India and other contemporary settings.[31]

To measure development, we use the 1911 population density of the district and the share of the population supported by commerce. Using information in the historical district gazetteers, we construct the 1911 value of income taxes collected

[26] See works listed under 'Official publications'.
[27] A district is an administrative unit below a state or province in India.
[28] Government of India, Singh, and Chandramouli, *Administrative Atlas*.
[29] Government of India, *Census of India*.
[30] The index includes Hindu castes that accounted for than 1% of the province's population in 1911, along with Muslims, Christians, Buddhists, Sikhs, Jains, tribal groups, and others including castes that did not constitute 1% of the province's population, and small religious groups.
[31] Alesina, Baqir, and Easterly, 'Public goods'; Miguel and Gugerty, 'Ethnic diversity'.

Figure 1. *Map of India, 1991*

in the district.[32] Income taxes in this period were generally collected from workers participating in the formal economy. They were levied on a small share of the population and capture the higher tail of the formal income distribution. These variables are intended to capture differences in demand for education across districts.

[32] Government of India, *India, District and Provincial Gazetteers*.

We also use the gazetteers to extract information on 1911 public education spending by the rural district boards and 1911 land tax revenues. These land revenues were an important determinant of public spending and offer a potential instrument for public education expenditures. The United Provinces (modern-day Uttar Pradesh) district gazetteers do not report the specific revenues collected, but instead report the revenue demand or the amount that districts were expected to collect in specific years between 1905 and 1911. The reported year varies by district. Since the revenue demand was fixed for 20 to 30 years, the inconsistency in the reporting year should not create significant problems. There could in principle be differences between revenue demand and revenue collected, but the regressions include geographic and economic controls to address problems arising from such differences.

Given the importance of geography for the computation of land revenues and education spending, we use the India agriculture and climate dataset prepared by the World Bank to construct indicator variables for black, red, and alluvial soil. We use the same dataset to extract information on altitude and latitude, and construct a dummy for coastal districts using the *Administrative Atlas*. Contemporary data on literacy and graduation are reported in the Indian district database compiled from the censuses of 1961 to 1991.[33] Using these data, we construct measures of total literacy, rural literacy, male rural literacy, rural literacy of those aged 10 to 19, rural primary school graduates, and rural secondary school graduates per head of rural population.[34] We focus on rural outcomes because our measure of historical spending was targeted at rural areas. We also obtain district-level information on the availability of primary schools, middle schools, and high schools in 1981 to assess the extent of public goods provision.[35]

In the Indian census, literacy is self-reported and defined as the ability to both read and write in any language.[36] Since both literacy and graduation are self-reported, it is unclear if one measure is preferable to the other. One concern with primary or secondary school graduates is whether individuals counted themselves as graduates based on examinations, a certain number of years of schooling, or mere attendance. On account of these differences, graduates may be a noisier measure of achievement than literacy. The correlation between primary school graduates per head of rural population and the different literacy measures ranges from 0.5 to 0.7 for 1961 and increases to 0.9 by 1991.

Tables 3a and 3b present summary statistics for the key variables. Table 3a focuses on literacy and graduates for the period 1961–91. In the first panel, we

[33] Vanneman and Barnes, 'Indian district data'; Government of India, Singh, and Chandramouli, *Administrative Atlas*; Sanghi, Kumar, and McKinsey, 'India climate and agriculture dataset'.

[34] The 1961 census reports only two types of rural graduates—primary school and matriculates (students who have passed the tenth grade exam, which is the final year of high school in India). From 1971 the census reports rural primary school, middle school, matriculates, non-technical diploma, diploma, and college graduates. In this case, secondary school graduates are the sum of the matriculates and other higher graduate categories. If an individual was a matriculate and diploma holder, she would be counted in both categories. This generates an upward bias to the proportion of secondary school graduates per head of rural population.

[35] Lakshmi Iyer provided us with these data, constructed from the 1981 census village directory data; India, *Census of India, 1981*.

[36] Our understanding is that the head of the household reports on each member's literacy. According to the Population Reference Bureau, 'a positive response is given if the individual has had any schooling at all. In India, the census enumerator is instructed, when there is doubt, to see whether a person listed as literate can read any part of the enumerator's manual and write a simple sentence. How often this is done in practice is not known'; Sharma and Haub, 'Examining literacy'.

Table 3a. *Summary statistics for education*

	Mean	*Std. dev.*	*25th percentile*	*Median*	*75th percentile*	*Min.*	*Max.*
1911							
Total literacy	5.0%	2.8%	3.0%	4.4%	5.8%	1.7%	15.4%
Male literacy	9.2%	5.0%	5.4%	8.2%	10.5%	3.2%	27.4%
Female literacy	0.7%	0.8%	0.3%	0.5%	0.8%	0.1%	6.4%
1961							
Total literacy	22.8%	7.2%	17.1%	21.5%	28.8%	9.6%	41.9%
Rural literacy	18.9%	5.8%	14.4%	17.5%	22.9%	7.9%	36.0%
Male rural literacy	29.6%	7.7%	23.8%	28.4%	34.9%	12.4%	47.9%
Female rural literacy	7.7%	4.6%	4.4%	6.1%	10.3%	1.7%	24.6%
Rural literacy, population aged 10–19	34.4%	10.2%	26.4%	32.9%	40.6%	15.2%	59.6%
Rural primary school graduates[a]	6.3%	3.9%	3.7%	4.9%	8.1%	1.7%	26.1%
Rural secondary school graduates[b]	2.2%	2.1%	0.9%	1.5%	2.5%	0.4%	13.3%
1971							
Total literacy	28.1%	8.9%	20.7%	26.7%	35.4%	12.2%	48.8%
Rural literacy	23.8%	7.5%	17.9%	22.2%	29.9%	10.7%	45.4%
Male rural literacy	34.5%	8.9%	28.1%	32.8%	41.8%	16.3%	56.1%
Female rural literacy	12.4%	6.9%	7.0%	9.9%	18.2%	2.9%	34.9%
Rural literacy, population aged 10–19	44.5%	12.8%	35.3%	42.1%	54.3%	22.0%	76.5%
Rural primary school graduates[a]	15.4%	6.1%	10.7%	14.0%	19.0%	5.8%	34.9%
Rural secondary school graduates[b]	5.5%	4.3%	3.1%	4.2%	6.2%	1.8%	31.8%
1981							
Total literacy	34.8%	10.2%	26.6%	33.5%	43.5%	15.6%	56.5%
Rural literacy	30.0%	9.1%	23.1%	27.2%	37.9%	13.4%	52.1%
Male rural literacy	41.8%	9.8%	34.6%	40.5%	49.7%	20.5%	63.9%
Female rural literacy	17.5%	9.4%	10.1%	14.4%	24.7%	3.4%	42.8%
Rural literacy, population aged 10–19	50.1%	13.5%	40.9%	48.0%	60.4%	24.7%	85.2%
Rural primary school graduates[a]	18.8%	6.5%	13.5%	17.5%	23.9%	7.9%	35.6%
Rural secondary school graduates[b]	11.1%	7.5%	6.5%	8.9%	13.4%	3.2%	59.6%
1991							
Total literacy	41.6%	11.1%	32.1%	40.8%	51.3%	19.5%	63.3%
Rural literacy	37.0%	10.4%	29.1%	35.5%	46.0%	16.4%	59.9%
Male rural literacy	48.2%	10.2%	41.5%	47.8%	55.1%	24.7%	69.9%
Female rural literacy	23.6%	11.1%	15.4%	20.7%	31.7%	6.2%	52.4%
Rural literacy, population aged 10–19	62.1%	14.8%	51.8%	60.9%	74.3%	30.1%	90.8%
Rural primary school graduates[a]	25.4%	8.0%	19.9%	23.7%	31.6%	8.2%	45.0%
Rural secondary school graduates[b]	17.7%	11.4%	11.2%	14.7%	19.7%	5.8%	86.4%

Notes: a Per head of rural population.
b Per head of rural population.
Sources: See section II.

report total literacy and literacy by gender in 1911 for our sample of districts to illustrate the differences between colonial and independent India. Although literacy levels were low in the decades immediately following independence at 23 per cent in 1961, this was a marked improvement on the colonial period when total literacy averaged 5 per cent.[37] Similar to 1911, there were large differences between rural male and female literacy in 1961, which continued until 1991 (male literacy averaged 48 per cent in 1991 compared to 24 per cent for females). On the bright side, younger generations have experienced some progress because the cohort of those aged 10 to 19 had higher literacy levels, especially by 1991.

[37] Literacy levels remained low throughout the colonial period. See Chaudhary, 'Caste, colonialism and schooling', for details.

Table 3b. *Summary statistics for control variables*

	Mean	*Std. dev.*	*25th percentile*	*Median*	*75th percentile*	*Min.*	*Max.*
Colonial investments (rupees, 1911)							
Education expenditures/rural population	0.07	0.05	0.04	0.06	0.08	0.02	0.43
Medical expenditures/rural population	0.03	0.03	0.01	0.02	0.03	0.00	0.28
Infrastructure expenditures/rural population	0.15	0.17	0.06	0.10	0.18	0.02	1.74
Land revenues/rural population	1.64	1.19	0.84	1.55	2.04	0.03	8.32
Geographic controls							
Coastal dummy	14.2%	35.0%	0	0	0	0	1
Black soil (dummy)	19.6%	39.8%	0	0	0	0	1
Red soil (dummy)	18.9%	39.3%	0	0	0	0	1
Alluvial soil (dummy)	56.1%	49.8%	0	0	1	0	1
Latitude	23	5	21	24	27	9	32
Altitude	377	172	279	363	445	82	1,124
Mean annual rainfall	1,318	504	1,046	1,257	1,546	346	4,324
Social controls (1911)							
Fraction Brahman	5.9%	4.3%	2.8%	4.5%	8.7%	0.5%	23.6%
Fraction lower castes	18.0%	6.7%	13.0%	17.7%	22.7%	3.6%	37.5%
Fraction Muslim	12.9%	12.3%	4.8%	9.1%	16.0%	0.5%	59.5%
Fraction tribes	3.2%	8.5%	0.0%	0.0%	1.5%	0.0%	59.8%
Fraction Christian	0.9%	2.0%	0.1%	0.2%	0.8%	0.0%	14.6%
Social fragmentation index	0.82	0.09	0.79	0.84	0.89	0.38	0.92
Development controls (1911)							
Income taxes/population	0.05	0.05	0.02	0.04	0.07	0.00	0.42
Population density	0.39	0.24	0.19	0.40	0.54	0.07	1.85
No. of observations	148	148	148	148	148	148	148

Sources: See section II.

Readers may be familiar with the pattern of regional differences in India—the southern and western states consistently outperformed the eastern and northern states across the different measures of literacy. In our sample, total literacy in the districts of Gujarat and Tamil Nadu averaged 32 per cent in 1961 compared to 18 per cent in the districts of Bihar and Uttar Pradesh. These differences persisted up to 1991, with Gujarat and Tamil Nadu averaging over 50 per cent compared to 32 per cent in Bihar and Uttar Pradesh. There was also substantial variation within states. For example, the 1961 literacy rate averaged 25 per cent in the districts of West Bengal, with a high of 37 per cent and a low of 14 per cent. Similarly, total literacy within Uttar Pradesh ranged from a low of 10 per cent to a high of 39 per cent in 1961. Such within-state differences persisted until 1991, despite an increase in total literacy.

Table 3b reports summary statistics for the historical and other control variables. As seen, colonial investments in education varied tremendously, averaging 0.07 rupees per head of rural population, but ranging from a low of 0.02 rupees to a high of 0.43 rupees. In figures 2 and 3 we plot the relationship between 1911 expenditures and rural literacy in the cohort aged 10 to 19 for 1961 and 1991 respectively. Despite a few outliers, the graphs reveal a strong positive relationship between historical spending and subsequent literacy. These graphs are striking but they miss the role of factors such as income that may also influence spending and literacy. Therefore we now turn to regression analysis to unpack the relationship between historical spending and contemporary outcomes.

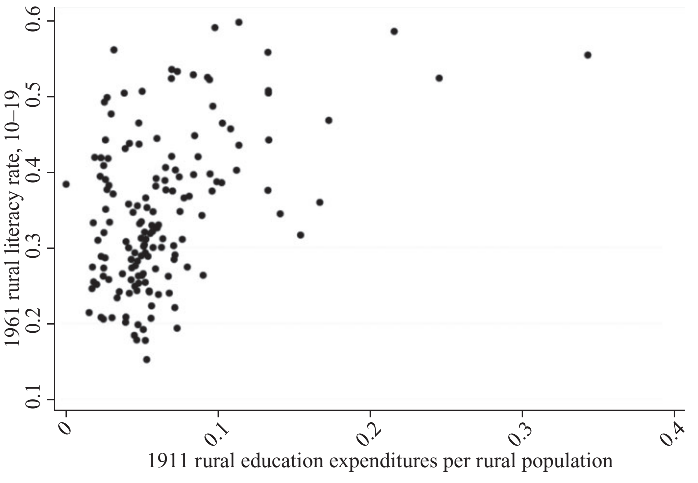

Figure 2. *Rural expenditures on primary education in 1911 and literacy in 1961*
Sources: See section II for details.

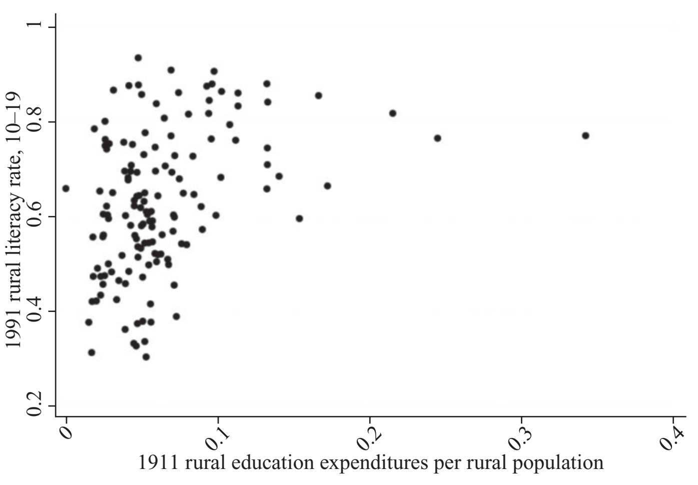

Figure 3. *Rural expenditures on primary education in 1911 and literacy in 1991*
Sources: See section II for details.

III

Empirical specification and econometric concerns

We begin the analysis by estimating the OLS relationship between post-independence literacy and 1911 colonial education investment, as shown below:

$$y_{is} = \alpha + \beta ColEducExp1911_{is} + \gamma X_{is} + \theta OtherColExp_{is} + \delta_s + \varepsilon_{is}$$

Here y_{is} is a measure of literacy in 1961, 1971, 1981, or 1991 in district i in state s. *ColEducExp*1911 represents public expenditures on rural education per head of rural population in 1911; X_{is} is a vector of geographic, social, and income controls; and *OtherColExp* is a vector of public expenditures on medical services and infrastructure in 1911. δ_s is a vector of state-specific fixed effects. As opposed to state fixed effects, we could have included province fixed effects for the historical British Indian provinces. However, we chose to exploit the variation within contemporary state borders because they were created during the 1956 linguistic reorganization of states, they represent linguistically similar regions, and they are unlikely to be correlated with the variation in colonial public investments. In a few cases, state borders are similar to the province borders (for example, Uttar Pradesh), but for the most part many more states were created than there were provinces, and some states include districts that were in different provinces.[38]

To identify the causal effect of colonial investments, we have to address the main endogeneity problem, that is, omitted variables bias. For example, if we fail to control for factors correlated with historical investment and post-1961 literacy, then we could mistakenly attribute the effect of an omitted variable to colonial investment. This problem arises because historical investments were not randomly assigned across districts. Colonial officials may have targeted more money to richer or more politically connected districts (in other words, those with a larger proportion of *brahmans* or other elite groups), which would bias our coefficients on colonial investments. As discussed in section II, the type of land settlement contributed to the regional variation because districts under the permanent settlement had their taxes fixed in cash for perpetuity and so their revenue base was constrained. Within regions, land tax revenues were an important source of variation in education spending because an additional tax on these revenues determined the level of revenues available for public investment. However, idiosyncratic factors also contributed to the revenues received by each district.

To ensure our OLS results for colonial investment are not picking up the effect of omitted factors that may influence subsequent literacy, we include three types of controls: geographic, social, and development and other historical investments. Geographic controls include indicators for red, black, and alluvial soil; altitude; and latitude. The social controls include the 1911 population share of the caste and religious groups mentioned in the data section and a district-level measure of caste and religious fragmentation. Finally, to control for differences in income and development across districts, we include historical measures of population density and income tax revenue per capita.

Public education spending in 1911 could also be related to public spending on local infrastructure and medical services that were managed by the district boards. Because land revenues were important to district board revenues, districts with higher land revenues may have outspent other districts in all categories of spending, namely primary education, local infrastructure, and medical services. Since the provision of the latter services could independently influence historical outcomes and in turn contemporary outcomes, we include 1911 public spending on infrastructure and medical services as controls. The public spending variables are standardized using the 1911 rural population.

[38] Our sample of British Indian districts includes those in the former provinces of Bengal, Bihar and Orissa, Bombay, Central Provinces, Madras, Punjab, and the United Provinces of Agra and Oudh.

OLS results

Table 4 presents the first set of results on literacy in 1961. Each number in the table is a coefficient from an OLS regression of the log of 1911 public education spending per head of rural population. Each column adds new controls. For example, column 1 is the raw correlation between literacy and colonial investments with no controls, column 2 adds state fixed effects, column 3 includes the geographic controls, column 4 adds the social composition controls, column 5 adds development controls, and finally column 6 adds other historical investments. Each panel focuses on a different measure of literacy (dependent variable).

In panel A, we see that total literacy is positively and strongly correlated with colonial investment. The coefficient is both economically and statistically significant, suggesting that a 10 per cent increase in historical spending translates into a 4.7 percentage point increase in literacy in 1961. This is a huge effect, given that in 1961 literacy averaged only 23 per cent. As we add more controls, we chip away at both the size and significance of the coefficient on historical spending. In the fully loaded specification in column 6, the coefficient on colonial investment suggests that a 10 per cent increase corresponds to a 2.2 percentage point increase in total literacy, but is not precisely estimated.

Panels B to F assess the effect of investment on other measures to improve our understanding of the mechanism from colonial spending to outcomes 40 years later. Unsurprisingly, we find that colonial investments had a larger impact on rural literacy, a finding that is robust across the different specifications. More interestingly, the findings in panel D suggest that historical investment is strongly correlated with literacy in the rural school age population, which suggests that colonial investment may be correlated with more recent public investment in education. Panel E, on primary school graduates, reinforces this interpretation. Again, we observe that a 10 per cent increase in investment in 1911 translates into a 2.7 percentage point increase in the number of rural primary school graduates per head of rural population.

In theory colonial investment should be unrelated to the number of secondary school graduates because it was historically targeted at primary education. So, in some sense, secondary school graduates offer a control outcome. While the coefficient is positive and significant for secondary school graduates in columns 1 to 3 (panel F), the magnitude drops significantly, along with statistical significance, once we add controls for income and development. The extensive controls appear to address the more obvious sources of endogeneity that would influence both primary and secondary education.

In panel A, we report the coefficients on other colonial investments, namely medical services and infrastructure. These coefficients are small in magnitude and statistically insignificant, suggesting that there are no serious cross-effects from colonial investment in other public services to education in 1961.[39]

In table 5, we extend the analysis to outcomes in 1971–91. We focus on the fully loaded OLS results corresponding to column 6 in table 3. The large and positive effects of colonial spending are apparent even as late as 1991, although the coefficients decrease in size by 1991. For example, the magnitude of the effects on

[39] In the interest of space we do not report the coefficients on these variables for the other outcomes because they were always small and statistically insignificant.

Table 4. *Do colonial education investments in 1911 have an impact upon post-independence outcomes, 1961?*

	No controls	*State fixed effects*	*Geographic controls*	*Social controls*	*Colonial development controls*	*Other colonial investment controls*
Panel A: Total literacy rate						
Ln education expenditures/rural population, 1911	0.0466***	0.0371***	0.0314***	0.0145*	0.0226**	0.0221
	[0.008]	[0.009]	[0.009]	[0.009]	[0.011]	[0.014]
Ln medical expenditures/rural population, 1911						−0.0028
						[0.010]
Ln infrastructure expenditures/rural population, 1911						0.0024
						[0.009]
Panel B: Rural literacy rate						
Ln education expenditures/rural population, 1911	0.0311***	0.0226***	0.0165**	0.0136*	0.0326***	0.0361***
	[0.007]	[0.007]	[0.007]	[0.008]	[0.010]	[0.013]
Panel C: Male rural literacy rate						
Ln education expenditures/rural population, 1911	0.0320***	0.0198*	0.0084	0.0070	0.0347**	0.0430**
	[0.010]	[0.011]	[0.011]	[0.012]	[0.016]	[0.019]
Panel D: Rural literacy rate, population aged 10–19						
Ln education expenditures/rural population, 1911	0.0734***	0.0496***	0.0423***	0.0389***	0.0606***	0.0767***
	[0.012]	[0.012]	[0.012]	[0.015]	[0.018]	[0.021]
Panel E: Rural primary school graduates per head of rural population						
Ln education expenditures/rural population, 1911	0.0352***	0.0299***	0.0310***	0.0303***	0.0263***	0.0268***
	[0.008]	[0.006]	[0.006]	[0.007]	[0.007]	[0.008]
Panel F: Rural secondary school graduates (or higher) per head of rural population						
Ln education expenditures/rural population, 1911	0.0081***	0.0092**	0.0086**	0.0012	−0.0037	−0.0072
	[0.003]	[0.004]	[0.004]	[0.003]	[0.004]	[0.005]
Observations	148	148	148	148	148	148
Controls						
Geographic	No	Yes	Yes	Yes	Yes	Yes
Social	No	No	Yes	Yes	Yes	Yes
Development	No	No	No	Yes	Yes	Yes
State fixed effects	No	No	No	No	Yes	Yes
Other colonial investments	No	No	No	No	No	Yes

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Geographic controls include dummies for red, black, and alluvial soil; altitude; latitude; and coastal districts. Social controls include the 1911 fraction of Brahmans, Muslims, Christians, lower castes, and tribes, and a caste and religious fragmentation index. Development controls include 1911 income taxes per capita, 1911 population density of the district, and 1911 population supported by commerce. Other investment controls include 1911 medical expenditure and infrastructure expenditure per head of rural population.
Sources: See section II.

Table 5. *Do colonial education investments have an impact upon post-independence literacy, 1971–91?*

	Literacy rate				*Rural primary school graduates per head of rural population*	*Rural secondary school graduates per head of rural population*
	Total	*Rural*	*Male rural*	*Rural, aged 10–19*		
Panel A: 1971						
Ln education expenditures/ rural population, 1911	0.0226	0.0392***	0.0449**	0.0714***	0.0317*	–0.0180*
	[0.015]	[0.014]	[0.018]	[0.022]	[0.017]	[0.010]
Panel B: 1981						
Ln education expenditures/ rural population, 1911	0.0313**	0.0511***	0.0557***	0.0793***	0.0340***	–0.0214
	[0.015]	[0.015]	[0.018]	[0.024]	[0.010]	[0.018]
Panel C: 1991						
Ln education expenditures/ rural population, 1911	0.0300*	0.0511***	0.0516***	0.0648**	0.0269*	–0.0474
	[0.018]	[0.018]	[0.019]	[0.027]	[0.014]	[0.030]
Observations	148	148	148	148	148	148
Controls						
Geographic	Yes	Yes	Yes	Yes	Yes	Yes
Social	Yes	Yes	Yes	Yes	Yes	Yes
Development	Yes	Yes	Yes	Yes	Yes	Yes
State fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Other colonial investments	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Geographic controls include dummies for red, black, and alluvial soil; altitude; latitude; and coastal districts. Social controls include the 1911 fraction of Brahmans, Muslims, Christians, lower castes, and tribes, and a caste and religious fragmentation index. Development controls include 1911 income taxes per capita, 1911 population density of the district, and 1911 population supported by commerce. Other investment controls include 1911 medical expenditures and infrastructure expenditures per head of rural population. Data on literacy rates for those aged 10 to 19, primary school graduates, and secondary school graduates are missing for 10 districts in 1991.
Sources: See section II.

rural school age literacy (ages 10–19) is remarkably similar from 1961 to 1981 and declines somewhat in 1991. Similar to table 4, we find no significant effect of historical spending on the number of secondary school graduates in 1971–91. Broadly, we find that colonial investment had far-reaching effects on post-independence Indian literacy, especially on rural school age literacy, which suggests a link between colonial investment, contemporary investment in education, and contemporary outcomes.

IV

Selection

Since colonial investment in education was not randomly assigned across districts, there is a concern about the potential endogeneity of this investment, despite the extensive set of controls included in the OLS regressions. As noted in section I, land revenues had a strong influence on the total amount of revenues available to the rural boards, but these revenues were then split among education, infrastructure, medical services, and other minor categories. In theory, the composition and preferences of the board would influence where they wanted to spend their money.

The historical context does not provide a clear indication of whether we might expect the OLS estimates to be positively or negatively biased. Districts with more alluvial soil and more favourable agricultural conditions may have generated more revenues, but then board chairmen and influential elites in these districts may have wanted to spend the money on infrastructure (for example, roads) to promote agriculture and trade as opposed to primary education.

To assess whether colonial investments on education were targeted in larger measure to more economically favourable districts (positive selection) or less economically favourable districts (negative selection), table 6 studies the correlation between colonial investment and observable differences in geography, development, and social composition across districts. Regressions 1 to 3 do not include any state fixed effects, while regressions 4 to 6 do include state fixed effects. Geography is an important determinant of historical investment across states, but not within states. Coastal districts, those with alluvial soil, and those at

Table 6. *Endogeneity of colonial investments*

	Log education expenditures per head of rural population, 1911					
	(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*
Coastal	0.3284*	0.1747	0.1734	0.1162	0.0319	0.0434
	[0.189]	[0.208]	[0.182]	[0.125]	[0.130]	[0.124]
Black soil	0.5557***	0.6143***	0.3875***	0.0735	0.1775	0.0871
	[0.164]	[0.162]	[0.138]	[0.128]	[0.109]	[0.115]
Red soil	–0.4665***	–0.5116***	–0.5539***	–0.0497	–0.0776	–0.1333
	[0.113]	[0.133]	[0.124]	[0.101]	[0.098]	[0.098]
Alluvial soil	0.1368	0.1566	0.1480	–0.0577	–0.0161	–0.0624
	[0.107]	[0.125]	[0.109]	[0.067]	[0.067]	[0.065]
Latitude	–0.0054	0.0142	0.0142	0.0076	0.0158	0.0117
	[0.013]	[0.015]	[0.014]	[0.022]	[0.023]	[0.023]
Altitude	0.0009***	0.0006**	0.0002	0.0001	–0.0000	–0.0000
	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
(measured in 1911)						
Fraction Brahman		0.6043	1.7983		–0.1080	0.0208
		[1.313]	[1.177]		[1.044]	[0.925]
Fraction lower castes		–1.1094	–0.8398		0.5011	0.5238
		[0.776]	[0.705]		[0.590]	[0.573]
Fraction Muslim		–0.8965	–0.7704		–0.6477	–0.8011*
		[0.564]	[0.569]		[0.421]	[0.405]
Fraction tribes		–1.3650**	–0.8557		–1.0907**	–0.7553*
		[0.546]	[0.532]		[0.418]	[0.403]
Fraction Christian		4.4304	1.2699		5.1147**	3.0787*
		[2.999]	[2.449]		[1.968]	[1.851]
Caste and religious fragmentation		–0.8005	–0.5198		–0.9294*	–1.1118**
		[0.585]	[0.506]		[0.497]	[0.510]
Population density			–0.9018***			–0.1471
			[0.188]			[0.132]
Fraction commerce			2.5272			3.2482*
			[2.064]			[1.647]
Log income taxes per capita			0.1919***			0.0839
			[0.064]			[0.059]
Observations	148	148	148	148	148	148
R^2	0.152	0.257	0.357	0.754	0.779	0.802
State fixed effects	No	No	No	Yes	Yes	Yes

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Sources: See section II.

higher altitudes tend to be positively correlated with rural investment in primary education. On the other hand, population density is negatively correlated with rural education spending. Since our measure of colonial investment was directed at rural areas, it is unsurprising that they are negatively related to population density, as urban areas were more densely populated than rural villages.

Richer districts outspent poorer districts on average given the large and significant coefficient on income taxes across the different specifications. We also find that districts with larger proportions of Christians had higher levels of spending, while districts with more tribal groups and a higher degree of caste and religious diversity had lower levels. The Christian effect may be related to missionary activities on both the education and proselytization front. However, this does not offer much guidance on selection because missionaries in colonial India often operated in poorer and less developed districts, which could offset the positive selection from the income effect. While these correlations are informative regarding the broad patterns of colonial investment, they do not offer precise guidance on the direction of the endogeneity bias, or whether the different correlations offset each other.

Instrumental variable results

We draw on Chaudhary to address the potential endogeneity of colonial investments using land revenues.[40] That article studies the relationship between rural education expenditures in 1911 and literacy in 1921, using 1911 land revenues as an instrument. We paraphrase the main argument here. Land revenues strongly influenced the amount of money that districts had available to spend on rural primary education, because additional taxes on these revenues were the major source of funding for district boards. However, the idiosyncrasies of individual land assessment officers influenced the exact amount of the tax revenue, creating quasi-random variation across districts within the same province.[41] Assessors adopted different and often subjective criteria to determine the tax amount. According to Baden-Powell 'with all these different methods, it is apt to be supported that, after all, Settlement is very much a matter of individual taste and opinion'.[42] More importantly, the process of assessment was updated infrequently, every 20 to 30 years, and was not particularly sensitive to changes in the income and development of the district. Chaudhary discusses the qualitative evidence that supports the credibility of using land revenues as an instrument for rural public education spending.[43]

The first panel (A) of table 7 presents the correlations between land revenues and rural investment on primary education. The coefficient on land revenues is large and statistically significant across the different specifications even after we include controls and state fixed effects (regression 6). While the coefficient is higher in specifications without state fixed effects, the coefficient when we include

[40] Chaudhary, 'Taxation'.
[41] Baden-Powell, *Land-systems*; Baden-Powell and Holderness, *Short account*; Chand, *Financial system*; idem, *Local finance*.
[42] Baden-Powell, *Land-systems*, vol. 1, p. 338.
[43] We urge the interested reader to turn to the relevant sections of Chaudhary, 'Taxation', for more information on land revenues and their credibility as an instrument.

Table 7. *Instrumental variable estimates of colonial investments on post-independence literacy, 1961–91*

Panel A: First stage, log education expenditures/rural population, 1911 (dependent variable)

Ln land revenues/rural population, 1911	0.4704***	0.5818***	0.5076***	0.2555***	0.3316***	0.2836***
	[0.077]	[0.064]	[0.067]	[0.071]	[0.066]	[0.070]
Geographic	Yes	Yes	Yes	Yes	Yes	Yes
Social	No	Yes	Yes	No	Yes	Yes
Development	No	No	Yes	No	No	Yes
State fixed effects	No	No	No	Yes	Yes	Yes

Panel B: Fraction of rural population engaged in commerce (dependent variable)

	1961	*1971*	*1981*	*1991*
Ln land revenues/rural population, 1911	0.0003	0.0002	–0.0005	–0.0010*
	[0.001]	[0.000]	[0.000]	[0.001]
Controls	All	All	All	All

Panel C: Instrumental variable estimates from second stage

	Literacy rate				*Rural primary school graduates per head of rural population*	*Rural secondary school graduates per head of rural population*
	Total	*Rural*	*Male rural*	*Rural, 10–19*		
Panel A: 1961						
Ln education expenditures/rural population, 1911	–0.0104	0.0596**	0.0563	0.1263**	0.0283	–0.0395**
	[0.033]	[0.028]	[0.040]	[0.052]	[0.019]	[0.019]
Panel B: 1971						
Ln education expenditures/rural population, 1911	–0.0152	0.0661**	0.0667	0.1259**	0.0878**	–0.0846**
	[0.037]	[0.033]	[0.043]	[0.062]	[0.035]	[0.040]
Panel C: 1981						
Ln education expenditures/rural population, 1911	–0.0164	0.0662*	0.0645	0.0920	0.0645**	–0.1083
	[0.046]	[0.039]	[0.047]	[0.062]	[0.028]	[0.066]
Panel D: 1991						
Ln education expenditures/rural population, 1911	–0.0295	0.0443	0.0385	0.0300	0.0346	–0.1662*
	[0.053]	[0.047]	[0.052]	[0.066]	[0.037]	[0.099]
Observations	148	148	148	148	148	148

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. First stage statistics: F-test on land revenues = 0, 14.75 (p-value = 0.0002), Cragg-Donald Wald F-statistic = 19.61 (weak identification test). Geographic controls include dummies for red, black and alluvial soil; altitude; latitude; and coastal districts. Social controls include the 1911 fraction of Brahmans, Muslims, Christians, lower castes, and tribes, and a caste and religious fragmentation index. Development controls include 1911 income taxes per capita, population density of the district, and 1911 population supported by commerce. Data on literacy rates for those aged 10 to 19, primary school graduates, and secondary school graduates is missing for 10 districts in 1991.
Sources: See section II.

state fixed effects and the controls is almost two-thirds of the size when we exclude state fixed effects (regression 6 compared to regression 3). A 10 per cent increase in land revenues translates into a 3 per cent increase in education spending.[44]

Land revenues are strongly correlated with education spending, but a valid instrument in our context also has to be unrelated to unobservable factors correlated with contemporary literacy. To assess whether 1911 land revenues are systematically related to contemporary factors that may influence the demand for education, we study the correlation between these revenues and the share of the rural population supported by commerce in 1961–91 in panel B. While the commercial population is correlated with contemporary literacy,[45] historical land tax revenues are not statistically correlated with the contemporary commercial population. The coefficient on land revenues is small and insignificant for 1961–81; it is small but negative and mildly significant in 1991.

Panel C of table 7 reports the instrumental variable results for 1961 to 1991. The instrumental variable coefficients on colonial investments are small and statistically insignificant for total literacy across the four cross-sections. They are significant for rural literacy up to 1971, marginally significant for 1981, and statistically insignificant for 1991. We find the results on rural literacy in the population aged 10 to 19 the most interesting. The instrumental variable coefficients are larger in magnitude than the OLS but only statistically significant up to 1971. They are smaller in magnitude and statistically insignificant for 1981 and 1991. By 1991, the magnitude of the instrumental variable coefficient is just half of the OLS.

Surprisingly, the instrumental variable estimates are insignificant for primary school graduates in 1961 but are positive and significant for the 1971 and 1981 cross-sections. We do not have a good sense of why the 1961 results are different from 1971, but it could be related to how primary school graduates were measured in the 1961 census. Similar to the OLS, the instrumental variable estimates do not have any effect on secondary school graduates. Taken together, the instrumental variable results suggest that colonial investment strongly influenced rural literacy in the school age population up to 1971, but was no longer significant by 1981. The mechanism linking colonial investment to post-independence outcomes is one of limited persistence. The increased spending of the 1970s was successful in breaking historical ties.

Using data on the provision of schools in 1981, we confirm that the availability of primary schools by 1981 was also statistically unrelated to colonial investment. Table 8 presents the results using the proportion of villages with primary schools, middle schools, and high schools in 1981 as the dependent variable. The OLS specifications include the same set of controls as in earlier tables, and we instrument for colonial investment using colonial land tax revenues. Our OLS estimates on colonial investment for primary schools and middle schools are large and

[44] In regressions not reported here, we also included a dummy for Temporary Settlement districts and type of land tenure system in place (specifically, the proportion of non-landlord tenures) from Banerjee and Iyer, 'History'. Neither this variable not the land tenure variable was statistically significant after controlling for state fixed effects because of limited intra-state variation.

[45] In the interest of space, these results are not provided here, but they are available from the authors upon request.

Table 8. *Did colonial investments influence the provision of schools in 1981?*

	Proportion of villages with:					
	Primary schools	*Middle schools*	*High schools*	*Primary schools*	*Middle schools*	*High schools*
Ln education expenditure/ rural population, 1911	0.070** [0.030]	0.064*** [0.022]	0.032** [0.015]	0.083 [0.059]	0.013 [0.038]	-0.016 [0.031]
Observations	135	130	122	135	130	122
R^2	0.788	0.857	0.764	0.788	0.849	0.737
Estimation	OLS	OLS	OLS	IV	IV	IV

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. These regressions include controls for geography, social characteristics, development, and state fixed effects. Geographic controls include dummies for red, black, and alluvial soil; altitude; latitude; and coastal districts. Social controls include the 1911 proportion of Brahmans, Muslims, Christians, lower castes, and tribes, and a caste and religious fragmentation index. Development controls include 1911 income taxes per capita and the population density of the district.
Sources: See section II.

positive. However, the instrumental variable estimates mirror our findings on literacy in table 7. Colonial investments are not significantly correlated with the provision of public goods in 1981.

The results suggest that reforms initiated after the 1968 National Education Policy were successful in putting districts on a different path for public goods provision, at least with regard to primary education. A natural question then follows: why was the Indian government unable to initiate such reforms sooner? Although public spending increased in the decades immediately after independence, we believe that expanding basic primary education was not a national priority. The Indian government was focused on developing heavy industry and, if anything, promoted technical education. In the 1950s and 1960s, historically disadvantaged groups such as the Scheduled Castes and Scheduled Tribes were not important political players, and were perhaps unable to demand greater access to primary schools.

The Indian government may have also been constrained by political constituencies that benefited from the inequities in public spending and that supported the former colonial distribution of public funds. Rajan outlines a theoretical model whereby historical inequalities in endowments, public investments, and opportunities can create political constituencies unwilling to support pro-poor (that is, pro-education) policies.[46] In our context, higher rural public spending in the colonial period perhaps reduced relative inequality in access to schooling, which in turn may have created political constituencies willing to support the expansion of primary education in post-independence India. In comparison, areas with low levels of colonial investment had greater inequality in access to education and were unable to develop constituencies in favour of reform without strong backing from the central government. This theoretical model accords with Banerjee and Somanathan's evidence on the positive correlation between the political power of Scheduled Castes and their improved access to public goods between 1971 and 1991.[47]

[46] Rajan, 'Rent preservation'.
[47] Banerjee and Somanathan, 'Political economy'.

V

This article has examined the effects of colonial investment on recent education outcomes using a unique dataset of 148 Indian districts. While OLS estimates suggest large and persistent effects of colonial investment on post-independence literacy, the instrumental variable estimates suggest that colonial investment influenced outcomes up to 1971, but by 1981 was not a significant force. Unlike other contexts, the historical legacy of colonial education policies in India was relatively short-lived because the government explicitly pledged to improve universal access to public goods in the 1970s and followed through on this pledge by increasing public spending on education. Thus it is apparent that good policy can alter the effects of history. We recognize that it may be easier for governments to build more schools than to change more entrenched historical policies, but our study does highlight the fact that more work needs to be done to understand why other countries have been unable to introduce such policies in contexts where persistence in public goods provision is long-lived and widespread.

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